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Design Report Concurrent Task Dispatcher

Two scheduling strategies were tested in this project: a baseline FIFO scheduler and an optimized scheduler. Both were evaluated using the same workload of 1000 tasks generated at 20ms intervals with a 70/30 IO-to-CPU distribution. Each task required 200ms of simulated execution time, and the dispatcher enforced both worker availability (maximum of 8 workers) and a strict CPU usage cap of 100%.

The FIFO scheduler processed tasks strictly in arrival order. While this approach was simple and predictable, it was less efficient under mixed workloads. CPU-intensive tasks near the front of the queue often delayed smaller IO tasks, even when those IO tasks could have fit within the remaining CPU budget. As a result, FIFO frequently left usable CPU capacity idle and reduced effective worker utilization.

The optimized scheduler improved this behavior by making scheduling decisions based on current CPU load. When CPU utilization was already high, the scheduler prioritized IO tasks because they required only 10% CPU and could more easily fit into the remaining capacity. When CPU usage was lower, the scheduler preferred CPU-heavy tasks

to increase utilization. This allowed the optimized scheduler to pack tasks more efficiently and reduce wasted CPU headroom.

In the FIFO experiment, the system achieved an average CPU usage of 89.65%, an average worker utilization of 5.07 active workers, and a total runtime of 39430ms. In the optimized experiment, the system achieved an average CPU usage of 96.45%, an average worker utilization of 5.46 active workers, and a total runtime of 36650ms.

The optimized scheduler outperformed FIFO in all measured categories. Average CPU utilization increased from 89.65% to 96.45%, showing that the optimized scheduler used available CPU resources more efficiently. Worker utilization also improved from 5.07 to 5.46 active workers on average, indicating that more workers remained productive throughout the simulation. Most importantly, total runtime decreased from 39430ms to 36650ms, reducing total execution time by 2780ms (about 7.1%).

Overall, the results show that FIFO is a valid baseline scheduling policy, but it performs less efficiently under mixed-resource workloads because it does not account for task cost. The optimized scheduler produced better throughput and resource utilization by making simple cost-aware dispatch decisions. This demonstrates that even small changes in scheduling policy can significantly improve concurrent system performance.